Pr Report

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Final Project Report — Programming Fundamentals Lab

University Name: National University of Computer and Emerging Sciences (FAST-NUCES)
Department: Department of Software Engineering (BSSE)
Course: Programming Fundamentals -Lab
Project Title: All In One Calculator
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Abstract

We have made a console based All In One Calculator in C programming language with various modes which include Scientific Mode, Unit Conversion Mode, Graphing Utility, and the starring feature, Ai Mode which responds to your queries. We used different types of algorithms to implement operator precedence based on Mathematics rules (e.g BODMAS). We used windows.h header file to run command prompt commands through code and implement an appealing console UI. We also used curl header file which is a windows built in program to send and fetch http requests through which we send requests to the Google provided Gemini API along with the user's queries (the user can also upload images) and send back response which is then converted from JSON to simple HTML and then opened in the browser as an HTML file. This project made us learn practical use of different algorithms and coding techniques and different C libraries. We also learned the use of APIs and interacting with them through C programming language.

1. Introduction

When we talked to different batch mates and looked around us, we found that the students were having trouble in studying Calculus because of the many disconnected apps they use like separate apps for simple calculator, scientific calculator, graphing calculator, Ai solver, etc. So we decided to change this and help all our batch mates and we made this All in One Calculator which allows you to use any type of calculator from scientific to Ai solver even the conversion calculator and graphing calculator. Now the user wont have to switch between different apps, making both studying and user experience great!

2. Objectives

- Provide better user experiance.
- Bring all of the different utilities to a single platform.
- Help all of the students.

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3. System Design

System Overview

The program runs in a loop, providing 4 different modes; 1. Scientific, 2. Conversion, 3. Graphing, 4. Ai Agent. The user can use any mode as many times as he wants and then get back to main menu to switch modes through a very easy command.

How the program flows:

Start → Main Menu → Pick a Mode (Scientific / Converter / Graphing / AI Agent) → Type Problems → See the Output → Go Back to Menu or Exit.

Algorithm (Scientific Mode)

1. Input: You type in something like 10 + 2 * 5. (We use safe input method by clearing the input buffer first.)

2. Conversion: The Shunting-yard algorithm converts from human notation (traditional maths notation) to Postfix notation (notation which is easy to understand for the computer).

4. Evaluation: The converted expression is then calculated and result is stored.

5. Output: You see your answer.

Input & Output

- Input: Math expressions, menu choices, queries for the AI, and image files.
- Output: Calculated answers, converted measurements, Graphs, and AI responses.

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4. Implementation

Language: C (C99 Standard)

Compiler/IDE: GCC / CLion / VS Code / Dev-C++

External Dependencies: windows.h, comdlg32, libcurl / System Calls

Key Features

- Parser-Based Calculator: This Calculator can easily Handle equations with brackets () and decimal points and calculate the result by following Mathematical rules.
- Graphing Mode: Plots x vs y coordinates dynamically right in the console.

- AI agent: Sends images and text to Google's Gemini 2.5 Flash model and fetches answers to your queries.
- Converter Mode: Provides you with a list of conversions to choose from and then converts your input to your desired unit.

Code Snippet (Sending request to AI Agent)

```
// command to send JSON data to Google Gemini API

sprintf(command,

  "curl -s -X POST
\"https://generativelanguage.googleapis.com/v1beta/models/gemini-2.5-
flash:generateContent?key=%s\" "

  "-H \"Content-Type: application/json\" "

  "-d @request.json > response.txt",

  API_KEY);


// Running the command via system call

int status = system(command);

if (status == 0) {

  // Open response.html and parse the JSON result

// ...

}
```

Sample Output (Graphing Mode)

```
==============================================================
==
 FAST CALCULATOR PROJECT - GRAPHING MODE
==============================================================
==
Enter function of x: x*x


Plotting: y = x*x
*    |     *
 *   |    *
  *  |   *
  ** | **
   ******
```

5. Testing & Results

We tested everything against different scenarios to make sure it's stable, especially when it comes to the order of mathematical operations and connecting to the internet.

We tested all the modes repeatedly using different scenarios. Everything worked fine and we refined the code by debugging small errors/bugs.

6. Conclusion, Limitations & References

Conclusion

This project not only demonstrates the practical use of programming techniques but also to think and develop solutions for the betterment of people. This project made it quite easy to study Calculus for students by assembling all different types of calculators into one single entity.

Limitations

- Internet access is required to use the Ai agent.
- Since we're drawing graphs with text characters (ASCII), the graph quality is low.
- Since we used windows.h for the console output customization, so the code won't work on Linux or MacOS without some changes.

Future Enhancements

- Add a Real GUI.
- Save History of each user's calculations.
- Improve graph quality through real UI and providing support for trigonometric functions in the graphing mode.

References

- Documentation: Google Gemini API Docs (AI Studio).
- Algorithm: Edsger Dijkstra's Shunting-yard Algorithm (Wikipedia).
- Book: The C Programming Language by Brian Kernighan and Dennis Ritchie.
- Resources: StackOverflow community for help with curl command syntax.

Pr Report

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